

Chapter 1

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1 The Lebesgue Integral

Problem 2. Check that if $f_1, f_2, \dots$ are real continuous functions on the line and if $f = \lim_{n \rightarrow \infty} f_n$ exists pointwise, then $(x : 0 \leq f(x) < 1)$ is measurable. Hint: $(x : f(x) < 1) = \bigcup_{k \geq 1} \bigcup_{m \geq 1} \bigcap_{n \geq m} (x : f(x) \leq 1 - \frac{1}{k})$

Proof. Since $f_1, f_2, \dots$ are real continuous functions on the line, they are Borel Measurable. Since f is seen as the pointwise limit of $f_1, f_2, \dots$, it is borel measureable since borel measurable functions are closed under countable pointwise limits. $\forall n \in \mathbb{N}$, $[1, \infty)$ is a closed sets, thus $f_n^* \{[1, \infty)\}$ is closed thus borel measurable. Thus $(x : f(x) < 1) = \bigcup_{k \geq 1} \bigcup_{m \geq 1} \bigcap_{n \geq m} (x : f(x) \leq 1 - \frac{1}{k})$ is borel measurable since measurable sets are closed under countable unions and intersections. $(x : 0 \leq f(x) < 1) = (x : 0 \leq f(x)) \cap (x : f(x) < 1)$ is measurable. $\square$

Problem 3.

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Calculate $\int_0^1 f$

Proof.

$$\begin{aligned} \int_0^1 f &= 1 * \mu(x \in [0, 1] : f(x) = 1) \\ &= 1 * \mu(x \in [0, 1] : x \in \mathbb{Q}) \\ &= 0 \end{aligned}$$

$\square$

Claim 4.

$$\lim_{n \rightarrow \infty} \int f_n = \int f$$

if $f_1 \geq f_2 \geq \dots$, $\lim_{n \rightarrow \infty} f_n = f$, and $\int f_1^+ < \infty$

Proof. Let $g_n(x) := f_1(x) - f_{n+1}(x)$. Thus $g_0 = 0$ and $0 \leq g_1 \leq g_2 \leq \dots$. Since $\lim g_n = f_1 - f$, $\lim \int g_n = \int(f_1 - f)$.

$$\begin{aligned}\lim \int g_n &= \int (f_1 - f) \\ \lim \int (f_1 - f_n) &= \int f_1 - \int f \\ &= \lim \int f_n = - \int f \\ \lim \int f_n &= \int f\end{aligned}$$

□

2 Geometry of $L^2(Q)$ continued

Problem 1.

Solution. Let $\{e_i\}_{i=1}^n$ be a family of unit perpendicular members.

$$\begin{aligned}
 |e_i - e_j|^2 &= \langle e_i - e_j, e_i - e_j \rangle \\
 &= \langle e_i, e_i - e_j \rangle - \langle e_j, e_i - e_j \rangle \\
 &= \langle e_i, e_i \rangle - 2\langle e_i, e_j \rangle + \langle e_j, e_j \rangle \\
 &= 1 - 2\langle e_i, e_j \rangle + 1 = 2
 \end{aligned}$$

Thus $|e_i - e_j| = \sqrt{2}$ ■

Problem 2.

Proof. $\forall c_n \in \mathbb{C}$ and e_n part of the family of unit perpendicular, $c_n e_n \in L^2(Q)$. Thus $f_n(x) := \sum_{k=1}^n c_k e_k \in L^2(Q)$. Take the sequence $\{f_n\}_{n \rightarrow \infty}$. $\forall \epsilon > 0$, for this $\epsilon \exists K \in \mathbb{N} \ni \forall n \geq K, \sum_{k=1}^n |c_k|^2 < \epsilon$. Thus $\forall m > n \geq K$

$$\begin{aligned}
 |f_m - f_n| &= \left| \sum_{k=1}^m c_k e_k - \sum_{k=1}^n c_k e_k \right| \\
 &= \left| \sum_{k=n+1}^m c_k e_k \right| \\
 &= \sqrt{\int_{-\infty}^{\infty} \left| \sum_{k=n+1}^m c_k e_k \right|^2 dx} \\
 &= \sum_{k=n+1}^m |c_k|^2 < \epsilon \quad \text{(Because } \sum_{k=n+1}^{\infty} |c_k|^2 \text{ is convergent)}
 \end{aligned}$$

Thus $\{f_n\}_{n \in \mathbb{N}}$ is cauchy thus convergent in L_2 . Thus $\exists f = \lim_{n \rightarrow \infty} f_n \in L_2$ pointwise.

$$\begin{aligned}
 \|f\|^2 &= \int_{-\infty}^{\infty} |f(x)|^2 dx \\
 &= \int_{-\infty}^{\infty} \left| \sum_{n=1}^{\infty} c_n e_n(x) \right|^2 dx \\
 &= \sum_{n=1}^{\infty} |c_n|^2 \\
 &= \sum_{n=1}^{\infty} c_n \overline{c_n} \\
 &= (c_n \cdot c_n)_{\mathbb{C}} \\
 &= |c|^2
 \end{aligned}$$

□

Problem 10. $\forall A \subseteq L^2(Q)$, $A^\perp$ is closed

Proof. Let $\{y_n\}_{n \in \mathbb{N}} \subset A^\perp$ such that $y_n \rightarrow y$.

$$\langle y, a \rangle = \lim_{n \rightarrow \infty} \langle y_n, a \rangle = 0 \implies y \in A^\perp$$

Thus $A^\perp$ is closed. □

Problem 4. Check that $f \in C^\infty(S^1) \iff \hat{f}$ is rapidly decreasing in the sense that $n^p \hat{f}(n)$ approaches 0 as $|n| \rightarrow \infty$ for every $p < \infty$

Proof. $\implies$: Suppose $f \in C^\infty(S^1)$.

Since,

$$\hat{f}'(n) = 2\pi i n \hat{f}(n)$$

thus $\forall p > 0$ since $f \in C^\infty(S^1)$ and $f^{(p)}$ is continuous:

$$f^{(\hat{p})}(n) = (2\pi i n)^p \hat{f}(n) \implies \hat{f} = \frac{f^{(\hat{p})}(n)}{(2\pi i n)^p} = \mathcal{O}(-p)$$

Thus $\forall p > 0$: $\sum_{-\infty}^{\infty} |\hat{f}' n^{p-2}|$ is convergent and $\hat{f}$ is rapidly decreasing.

$\Leftarrow$: Suppose $\hat{f}$ is rapidly decreasing. $f(x) = \sum \hat{f} e_n(x)$. Thus $f'(x) = \sum \hat{f}' e_n'(x)$. By the given hint, $f'(x) = \sum \hat{f}' e_n'(x) = \sum \hat{f}' 2\pi i n e^{2\pi i n x} = 2\pi i \sum n \hat{f}' e_n$ converges uniformly to a periodic function f_1 . Thus f_1 exists. Since $\hat{f}$ decreases rapidly, $n \hat{f}$ decreases rapidly. Now apply induction. Thus $f \in C^\infty(S^1)$ □

Problem 5. Check that the arithmetic means $n^{-1}(x_1 + \dots + x_n) \rightarrow y$ if $x_n \rightarrow y$.

Proof. $\forall \epsilon > 0$, for this $\epsilon \exists K \in \mathbb{N} \ni \forall n \geq K, |x_n - y| < \epsilon$. Moreover $M = \sup(x_n - y)$.

$\exists L \in \mathbb{N} \ni \forall n \geq L, \frac{K}{n}(M - \epsilon) < \frac{\epsilon}{2}$. Take $A = \max(K, L) < 1$

$$\begin{aligned}
\left| \frac{1}{n} \sum_{k=1}^n x_k - y \right| &= \left| \frac{1}{n} \sum_{k=1}^n (x_k - y) \right| \\
&\leq \frac{1}{n} \sum_{k=1}^n |x_k - y| \\
&= \frac{1}{n} \sum_{k=1}^K |x_k - y| + \frac{1}{n} \sum_{k=K+1}^n |x_k - y| \\
&< \frac{1}{n} \sum_{k=1}^K |x_k - y| + \frac{1}{n} \epsilon \sum_{k=K+1}^n 1 \\
&= \frac{1}{n} \sum_{k=1}^K |x_k - y| + \frac{n-K}{n} \epsilon \\
&< \frac{K}{n} M + \frac{n-K}{n} \epsilon \\
&= \frac{K}{n} (M - \epsilon) + \epsilon < 2\epsilon
\end{aligned}$$

□